

php → Youtube

ફે ક્યુર - (કિય)

<? php

showing something :

<? php

echo "□";

must use
કોઈ સિસ્ટમ
કોઈ

→ એક show
કરવો.

આવશ્યક

<? php

echo "□";

echo "□";

→ એક લાઈન માત્ર બતાવે.

?>

echo "□
";

echo "□";

→ બાજુમાં
લાઈન
બતાવે

→ variable set :

\$□ = "□";

→ યા store કરાવવા.

\$name = "Eharon";

echo \$name; → var print

wrong [কিছন: কুছন letter থাকা পাছা]

\$1 name

include use:

একটা file.php

<?php

echo "File One";

?>

Required statement:

<?php

require 'file.php';

?>

→ same কাজ করে include
এবং আরও

আরেকটা file

<?php

include 'file.php';

?>

output = File
One

অর্থঃ oi file
এবং সব চলে আসবে

include:

include '□' ; → file

বা প্রান্ত

error msg

দেখানো দিবে

পরের অংকগুলো
show করবে

→ continue execution

→ Error type : warning

Required:

require '□' ;



পরের অংক

কাজ করবে না।

error msg দেখানো
করবে।

→ stop execution

→ Error type : fatal
error

session

Get & Post:

`<form action = "process.php" method = "get">`

~~<input~~

User_name: `<input type="text" name="username"><br>`

Password: `<input type="text" name="pass"><br>`

`<input type="submit" value="Log in">`

`</form>`



User_name:

Password:

→ php to get method apply:

`<? php`

`if (isset($_GET['username']) && isset($_GET['pass'])) {`

`$username = $_GET['username'];`

`$password = $_GET['pass'];`

variable & store करेगा,

check
NULL

if (\$username == 'saud' & \$password == '123')

echo 'Log in successful';

} else {

echo 'Error Log in';

login ની રજા
લેવા દેખાડે.

!>

વિનુ અભાગી રજા → url & name & pass
show કરાવે.

POST method apply:

get/GET ના જગ્યાએ post/POST replace કરાવે

POST method apply રજા.

તમારે url & name, password (દેખાવે).

if (!empty(\$username)) → વ્યવસ્થા કરી તારનાં
check કરાવે દેખાડે.

GET

1. Data appears in the URL
2. Less secure
3. Limited (URL length limit)

Best for → search forms,

Filtering, when data is not sensitive

POST

1. data appears in the req. body.
2. More secure.
3. No practical size limit.

Best For - Login forms, Registration forms, password.

Data is sent inside the HTTP request body. not visible in the ~~visible in the~~ URL

→ Prevent SQL Injection in PHP.

*. Bind parameter:

```
$con = new mysqli('localhost', 'root', '', 'test');  
$stmt = $con → prepare("insert into  
products (sku, name, price) values(?, ?, ?)");
```

↓ table name ↳ row value set.

```
$stmt → bind_param('ssd', $x, $y, $z);
```

↓
string = sku value
string = name value
d = decimal value for price.

```
$x = "S44";  
$y = "Sh";  
$z = 42;
```

```
$stmt → execute();  
$x = "S5";  
$y = "S";  
$z = 40;  
$stmt → execute();
```

↗ এই statement
ব্যবহার করে
বৈচিত্র্যিক value
set করা (ওহি)।

POST দিয়ে form create করে ~~এই~~ form এ উই block
 নিয়ে \$x, \$y, \$z এ store করতে পারি \$- POST
 করতে পারে।

<form action="" method="POST">

<fieldset>

sku: <input type="text" name="sku">

Name: <input type="text" name="name">

Price: <input type="text" name="price">

</fieldset>



<input type="submit" value="submit">

connection

\$stmt query

bind parameters

\$x = \$_POST['sku'];

\$y = \$_POST['name'];

\$z = \$_POST['price'];

\$stmt execute();

উই

a safe way to insert variables into SQL queries using prepared statements.

*. Display data from my SQL table

products:

id	sku	name	price

table to data
show

<? php

```
include "connect.php";
```

database to connect

```
$sql = "select * from products";
```

→ no info.

```
$result = mysqli_query($con, $sql);
```

→ query execute

```
while($row = mysqli_fetch_array($result)) {
```

→ array is result

```
    print $row['sku'] . " " . $row['name'] . " " .  
    $row['price'];
```

```
    print "<br/>"; → new line.
```

associative
array

* HTML form (2) MySQL database & data insert using php;

→ post method.

<?php>

include (connection.php);

if (isset(\$_POST['cname']) && isset(\$_POST['contact']) && isset(\$_POST['address']))

\$cname = \$_POST['cname'];
\$contact = \$_POST['contact'];
\$add = \$_POST['address'];

\$sql = "insert into ^{table name} customer (Name, Phone, Address) values ('\$cname', '\$contact', '\$add')";

if (mysqli_query(\$conn, \$sql)) {

echo "inserted";

else {
echo "not inserted";

form create নিচে শো।

Name: <input type="text"/>

Contact: <input type="text"/>

Addr: <input type="text"/>

*. Update on edit query:

<?php

→ (যখানে connection দিচ্ছি)

require ('connection.php');

\$sql = "UPDATE article SET category =

→ এই column update করছি।

↓
table name

'national' WHERE id=4";

৬-দ্বাঃ যুক্ত করে একাধিক একসাথে করা যাবে

\$query = \$conn → query(\$sql);

\$query = mysqli_query(\$conn, \$sql); → query run করছে

if (\$query) {

echo 'Updated';

} else {

echo 'Not updated';

}

?>

Delete :

\$sql = "Delete from article where
id = 3";

↑ table name

\$query = ~~mysql~~ \$conn → query(\$sql);

* Select Query:

\$sql = "Select email, firstname From users";

↑ select column
show rows

↓
table
name

\$result = mysqli_query(\$conn, \$sql);

if (\$result) {

while(\$data = mysqli_fetch_array(\$result)) {

\$email = \$data['email'];

\$first name = \$da

echo '\$f', '\$email';



Select * from users ORDER BY (id) DESC

id to descending
order → show

* Session: is a way to store user info. on the server temporarily so that the data can be used across multiple pages of a website. Stores like (username, email, user id, cart items)

* Session: is a server side mechanism used to store user data temporarily across multiple pages.

Example file:

<?php

session_start();

\$_SESSION['username'] = 'Sakib';

?>

<?php

session_start();

echo \$_SESSION['username'];

?>

→ Sakib (name)

Example file:

<?php

session_start();

session_unset();

session_destroy();

?>

} Session data
clean.

How session works: It creates a unique session id, stores data on the server, and uses that id.

* Cookies:

<?php

setcookie(

time

* Session vs cookie:

Stored on server,
More Secure,
Temporary

Stored on browser
Less secure
can be long-term

* php code reuse:

(i) include: allows to reuse code from another php file.

(ii) Require: Works like include, but it stops execution if the file is missing.

(iii) include_once/require_once:

(iv) Using function: Create reusable functions inside a file.

func.php:

<?php

Function greet(\$name) {

name of the function

return "Hello", \$name; }

?>

index.php :-

<?php

include 'func.php';

echo greet("John");

?>

* View the form with php:

<?php

if(\$_SERVER["REQUEST_METHOD"] == "POST") {

\$username = \$_POST["username"];

\$email = \$_POST["email"];

echo "<h2> submitted info </h2>";

echo "Username: ", htmlspecialchars(\$username), "
";


```
echo "Email: " . htmlspecialchars($email);  
}
```

?>

* Purpose of mysql_connect() :-

It opens a connection to a MySQL database.

It requires four parameters:-

host name, username, password & database

((group 1) password = name).

```
$conn = mysql_connect("localhost", "root", "", "Test");
```

* Start a Session:

Session store user data (like login info) across multiple pages of a website.

* Table view:

<?php

connect

\$sql = "Select * From std";

\$query = mysqli_query(\$conn, \$sql);

echo "<table border='1'>";

echo "<tr><th>ID</th><th>Name</th><th>Dept</th>";

while(\$row = mysqli_fetch_array(\$query))

echo "<tr>";

echo "<td>". \$row['id']. "</td>";

echo "<td>". \$row['name']. "</td>";

echo

echo "</table>";

echo "</table>";

mysqli_close(\$conn);

?>

Create dropdown menu: in html

<h2> Select your Favorite Fruit </h2>

<select name = "fruits">

<option value = "apple"> Apple </option>

<option value = "banana"> Banana </option>

<option value = "grape"> Grape </option>

</select>

*. hyper link create that will open another tab:

. Select COUNT() AS total - employees from employees.

↓
count total employee.

→ Select MAX(salary) As max-salary
FROM employee;

→ Select * From employee where name
like 'A%';

↳ A দিয়ে শুরু

→ Select AVG(salary)

→ no. of employees in each dpt.

Select department COUNT(*) As
total employees From employee

GROUP BY department

→ BETWEEN → 4000 to 7000

↓
range বা range condition
রANGE

→ Select the second highest salary;

Select MAX(salary) As Second-highest
FROM employee where salary < (select

MAX(salary) FROM employee);

→ Create employee table:

CREATE Table Employee (

id INT primary key;

name varchar(50) not null,

dept varchar(50),

Salary Decimal(10,2),

age int

);

From Ai

Manifests : is a configuration file that contains essential information about ^{the} your app.

It defines -

- App name and package name.
- Permissions (e.g., internet, camera)
- Activities (screens of the app)
- App components and settings

The manifest file tells the android system what your app does, what it needs, and which components it has.

Before the app runs, Android reads this file first.

How it works?

Ans:

1. Android reads the manifest file first.
2. Checks required permissions
3. Identifies which activity to launch

4. Then runs the app.

* Resources: Resources are the non-code files of an Android application that are used to design the user interface and support the app.

They include -

→ Layouts (UI Design)

→ Images (drawable)

→ Strings (text)

→ Colors, styles, etc.

These are stored in the res folder.

Ex -

→ res/layout - XML design files

→ res/drawable - images

→ res/values - string, colors

* Gradle in Android: It is a build automation tool used to build, manage, and configure an Android project.

It handles -

→ ~~anti~~ compiling the code

→ Manage dependencies (libraries)

→ Creating the APK file.

* Activity: is a single screen of an Android application that the user can interact with

Each screen of an app (like login screen, home screen, profile screen) is called an activity.

Role of Activity: An activity works as a controller in Android. It:

→ Displays the UI using XML Layout

→ Handles user interactions (clicks, typing, etc.)

→ Connects UI (XML) with application logic (Java/Kotlin)

→ Controls what happens on a particular screen.

[It acts as a bridge between UI and logic]

How Activity works (Lifecycle): An activity goes through different states called lifecycle:

→ `onCreate()`: activity is created & UI is initialized.

→ `onStart()`: Activity becomes visible to the user.

→ `onResume()`: User can interact with the activity.

→ `onPause()`: Activity is partially hidden.

→ `onStop()`: Activity is no longer visible.

→ `onDestroy()`: Activity is destroyed & removed from memory.

Methods in Activity

`onCreate()` → Initialize activity & set UI

`setContentview()` → Attach XML layout to activity.

`findViewById()` → connect UI elements with code.

`Intent()` → Move from one acti

another.

onStart(), onResume() → Make activity interactive

onPause(), onStop() → Pause or stop activity

onDestroy() → clean up resources

Simple idea

XML = UI (what user sees)

Activity = Controller (handles UI +

logic connection)

Java/Kotlin = logic (what app does)

* Java class create: is a template for creating objects

```
class ClassName {
```

// variables

// methods

class name should start with a capital letter;

can contain letters, digits, -, and \$; No

space allowed, can't start with number,

use Camel Case (Recommended)

* Method in Java: is a block of code that performs a specific task.

```
void showData() {  
    // parameter  
    // system → print  
}
```

Rule: name start with lowercase letter, use camelCase, can't be a Java keyword,

must have return type (void, int, etc),
can take parameter or empty.

* Constructor: is a special method used to initialize objects.

```
className() {  
    // initialization code  
}
```

Rule: name must be same as class name,
No return type,

public class MainActivity extends AppCompatActivity

↓
is a class, it inherits features of
android activity App

#

AppCompatActivity allows support
for modern Android features
in older versions.

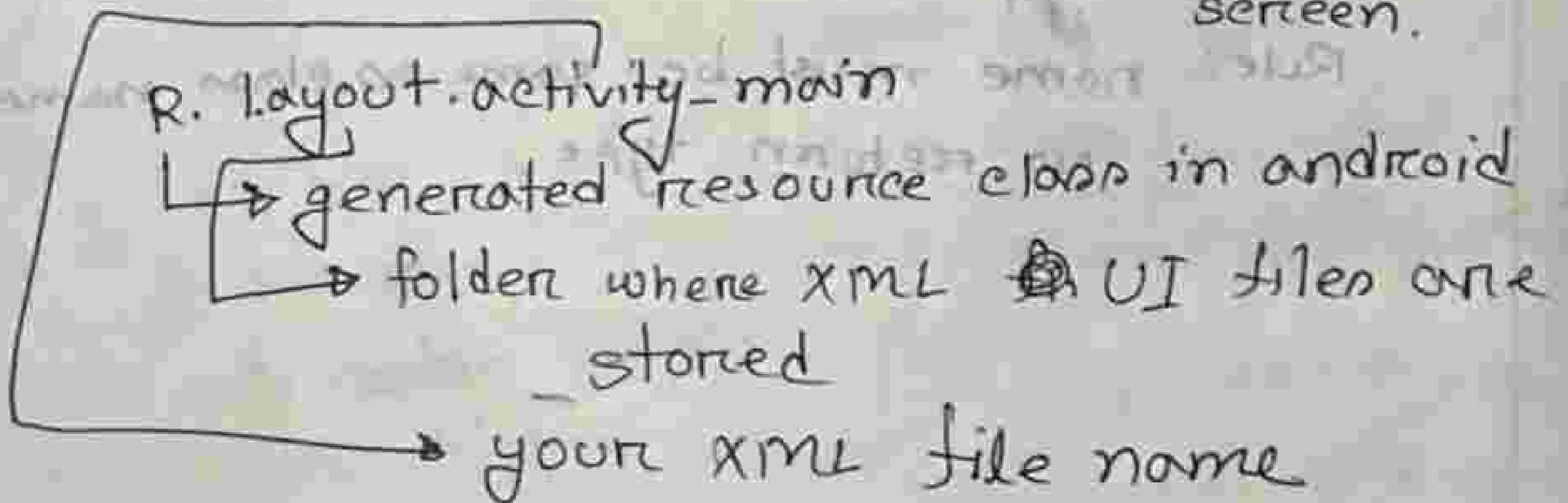
[This is your main screen of the
app.]

setContentView(R.layout.activity_main)

↳ This line is used to display the UI (XML
layout) on the screen inside an Activity.

It connects Java/Kotlin Activity with
your XML design file.

setContentView() → show this layout on
screen.



[It loads the XML layout file into the Activity
Screen]

Activity switching: means moving from one screen (activity) to another screen in an Android App. This is done using an Intent.

Intent intent = new Intent(this, SecondActivity.class);
startActivity() → execute the move

```
graph TD; Intent --> class; new --> object; this --> current_class[current class]; SecondActivity.class --> destination_class[destination class];
```

→ It is calling the Intent constructor.

Intent is called a messaging object because it ~~acts as a~~ works like a message carrier between Android components.

An Intent does not perform actions itself. Instead, it:

- carries information
- describes an action
- passes data or instruction

[it works like a messenger delivering instructions]

* onCreate() :

→ manual binding

findViewById() → is a method used to find and connect a UI element from XML to Java/Kotlin code.

How it work -

Each UI element in XML has an id. findViewById() use is that id to get the element.

automatic binding

View Binding → is a modern & safer way to connect XML UI with code without using findViewById().

* onResume() :

This is the state where the user can see and interact with the app.

→ Youtube (Anisul)

<LinearLayout - - -

android:layout_width="match_parent"

android:layout_height="match_parent"

<Button

android:layout_weight="1"

android:text="button1"

android:layout_width="match_parent"

android:layout_height="wrap^{odp}~~content~~"

>

<Button

android:layout_weight="1"

android:text="button2"

android:layout_width="match_parent"

android:layout_height="odp"

>

}

এক প্রকারে weight

আমি কখনে display কে ডিটাইল

সহজভাবে সমান করে দিচ্ছি।

match parent $\rightarrow$ parent এর সমান size হবে ডায়ালি screen
wrap_content $\rightarrow$ যতটুকু ডায়ালি লাগে ততটুকুই লিখে।

Linear layout

orientation = "vertical"

orientation = "vertical"
android:weightSum = "6" → full screen क
भरान 6 डाटा
डाटा करण.

< Button

android:layout_weight="2"

```
text = "button-1"
```

11 : Layout-width = "match-parent"

"L" : " - height = "odp"

17

< Button

Layout weight = 1

17

↳ Button

Layout - weight $2^0 1^0$

এবং,

Button Button1 = 2 জায়া

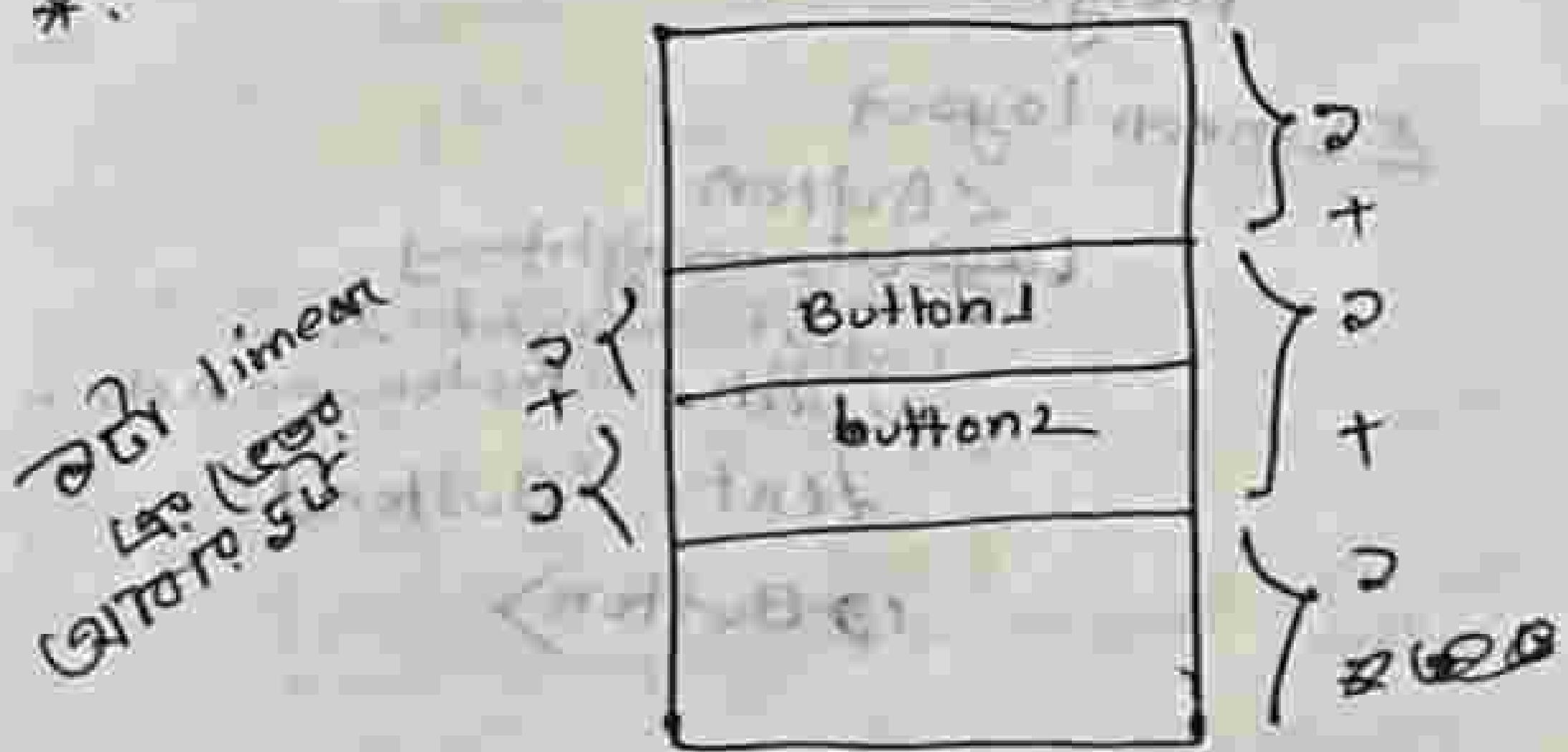
বাকি 4টা button 2, 3, 4, 5 = 1 জায়া

কম নিচে

সম্মত
6 জায়া
তার sum
= 6

যিনি যিনি device এ size এর পরিমিত (যদি সমান থাকে),
এইজন weight use করি।

*



এটা display / Linear
Layout এর মোট ৩টা
linear layout
নিহি।

< LinearLayout

android:layout_width = "match parent"

" " - height = " " "

orientation = "vertical"

weightSum = "3" → ৩টা জায়া সমান

< LinearLayout

android:layout_weight = "1"

android:layout_width = "match-parent"

height =

<LinearLayout>

<LinearLayout

layout_weight = "1"

width = "match-parent"

height =

<LinearLayout

<Button

layout_weight = 1

layout_weight = 2

width = "match-parent"

text = "Button 1"

</Button>

<Button

layout_weight = 1

</Button>

<LinearLayout

layout_weight = 1

width = "match-parent"

height =

</LinearLayout>

LinearLayout

→ একটি Layout কে হোরি ভাবে
Layout নিয়ে আলি display
করা হবে

Clean			←
7	8	9	+
4	5	6	-
1	2	3	x
0	=		÷

<Layout

weight-sum = 7

<TextView weight = 2

<Layout

weight sum = 4

orientation = horizontal

<Button

text = "Clean"
weight = "3"

</Button>

<Button

text = "←"

weight = "1"

</Button>

Sozzadul
Islam

exit
Java

EditText editText1, editText2;
TextView resultTV;

@ override

setContentView(R.layout.activity_main);

editText1 = findViewById(R.id.editText1);

editText2 = findViewById(R.id.editText2);

resultTV = findViewById(R.id.resultTV);

public void add(View view) {

String number1 = editText1.getText().toString();

String number2 = editText2.getText().toString();

String to
double

double result = Double.parseDouble(number1) +
Double.parseDouble(number2);
resultTV.setText(String.valueOf(result)); }

MySQL: are slower for real-time communication,
requires repeated querying

(Anisul)

* firebase :

what is firebase?

firebase is a remote database. Stores images, messages, user data etc. for your app. Easier, faster owned by google. Firebase manages all data real-time in the database.

benefits:

→ Create Application without backend server
→ No need extra money spent for backend server.

→ Sync real time data in the application

→ Quick display data in the application.

→ Faster than any backend web services.

→ No SQL database so it is more faster.

→ push notification

→ Analytics

→ Cloud storage

→ Dynamic linking

→ Real time updates happen instantly

→ No manual refresh needed

→ Cloud-based and scalable

→ Faster Synchronization between devices

→ Better for live sensor monitoring.

CodeItto Bro

(100m)

*. Firestore connection:

DatabaseReference ~~com~~ = FirebaseDatabase.getInstance().
getReference()

*. Code:

```
public class MainActivity extends AppCompatActivity
```

```
    EditText editText;
```

```
    Button button;
```

```
    DatabaseReference databaseReference;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState)
```

```
    {  
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        editText = findViewById(R.id.edit_text);
```

```
        button = findViewById(R.id.button);
```

```
        databaseReference = FirebaseDatabase.  
            getInstance().getReference();
```



```
button.setOnClickListener(new view.OnClickListener() {
```

@Override

```
public void onClick(View v) {
```

```
String data = editText.getText().toString();
```

```
databaseReference.setValue(data);
```

data को
store करेगा

```
});
```

```
databaseReference = FirebaseDatabase.getInstance().getReference().child("pathString: "myDB");
```

→ इसे use करके database में value

store करे-
जैसे input दिया Hello

myDB: Hello

जैसे H input दिया,

myDB: H

[Hello याद रखें यहाँ]

যদি চাই previous message এর output ছুঁতে Button
এ দেখতে চাই -

```
button2.setOnClickListener(new View.OnClickListener() {
```

```
public void onClick(View v) {
```

ICF
Fogsal

Member

key	name: E
	Age: 10
key	name: H
	Age: 9
	name: G
	Age: 8

unique রূপ।

এবং এর upload করে
চাইলে,

```
button.setOnClickListener(new View.OnClickListener() {
```

@Override

```
public void onClick(View v) {
```

```
String name = editText1.getText().toString()
```

```
String age = editText2.getText().toString()
```

```
String key = databaseReference.push().  
getKey();
```

Firestore conn.
getReference (path: "Member")

```
Member member = new Member(name,  
age);
```


databaseReference.child(key).setValue(member);

}

}

};

<EditText

android:hint = "Enter ~~you~~ name"

id = "@+id/nameEditTextId"

layout_width = "match parent"

<EditText

android:layout_marginTop = "15dp"

android:hint = "Enter Age"

id = "@+id/ageEditTextId"

layout_width =

layout_height =

<Button

text = "save data"

id = "@+id/saveDataButtonId"

layout_width = "match parent"

layout_height =

/LinearLayout>

Enter name
Enter Age
Save Data

Main.Activity.java

Button button;

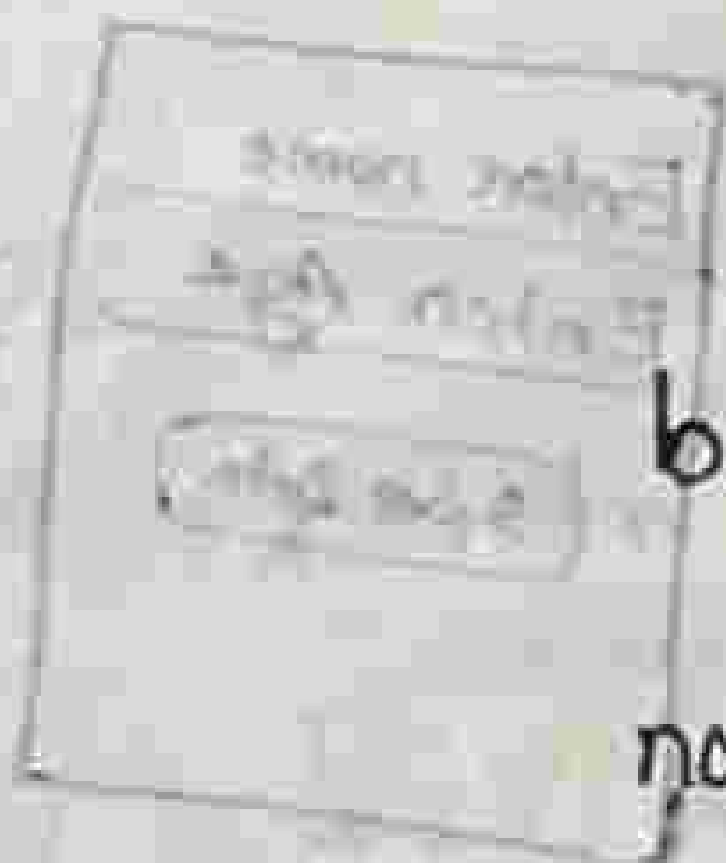
EditText nameEditText, ageEditText;

@Override

protected void onCreate(Bundle savedInstanceState)

super.onCreate(savedInstanceState);

setContentView(R.layout.activity_main);



button = findViewById(R.id.saveDataButtonId);

nameEditText = findViewById(R.id.nameEditTextId);

ageEditText = findViewById(R.id.ageEditTextId);

Database Ref add.

button.setOnClickListener(new View.OnClickListener() {

@Override

public void onClick(View v) {

String name = nameEditText.getText().toString();

String age = ageEditText.getText().toString();

String key = "previous";

*. Data fetch / Retrieve:

```
databaseReference.addValueEventListener(new ValueEventListener() {  
    @Override  
    public void onDataChange(DataSnapshot snapshot) {  
        for (DataSnapshot data: snapshot.getChildren()) {  
            String name = data.child("name").getValue(String.  
                class);  
            String age = data.child("age").getValue(String.  
                class);  
            System.out.println(name + " - " + age);  
        }  
    }  
});
```

*. Normal Text view → code PC → दर्शा

b.setOnClickListener

```
TextView dis = findViewById(R.id.text_display);  
String s = dis.getText().toString();  
dis.setText(s);
```


Youtube

Firestore Connect:

Android Studio ত project create →

MainActivity.java এর প্রথম line এ package name দেবে যা firebase connect এর

নাম package name এর অন্তর্ → nickname
যা মুক্তি দিয়ে। → json file download

করে Android Studio ত project view এ
সিমে App এ right click করে json file
টাকে paste করবে (google-services.json)।

→ Genovy (build.gradle) → plugins
Android
~~project~~ view এ build.gradle এ paste

↓
build.gradle (project: - -)
" (module: app)

Firestore এ সিমে Realtime database create করবে।

* Hacking : (unauthorized access)

bootforce attack → किस pass पढ़ेंगे
मगर assume करेंगे

SQL injection

what is google Doc

ideone.com/7bhfee → link

Apache → Xamp server & on

step

→ C drive → Xamp →htdocs & project
create

→ vscode & open code copy paste

form & action → login.php

→ login.php & code copy paste

→ SQL
[part by part code copy & build]

comment करें # फिर SQL &

select * from users where 1=1
↳ 1=true

SQL injection: SQL এর bypass করে (যা),

\$SQL = 'select * from users where user_id = \$
username and password = \$password';

1=1
user id এর বারানত → 'OR 1=1#
বাক্য পরে এর কন্ট্রল 2য়
এবং ~~access~~ করে।

DORK

inurl: login.php
site: .in → in → in dia/.bd → Bangladesh .pk → Pakistan

filetype: .pdf

open source intelligence
একটি উন্মুক্ত সূত্র

out of the
syllabus:

Wayback
machine
এ জায়গায়
site link like
kiu.ac.bd
search দিন
previous এ
site kemon
হিসাব করে SS
দেখা যায়

আমরা কেউ web
site পরিদর্শন
করলেও তার
previous
সুপার দেখা যায়
port পরিদর্শন
করে (ফরেনসিক)

inurl: login.php site: in

→ এতে মিলে indiana

indiana login.ph

site দেখা যায়

user_id = \$user_id

→ int. value
আসবে

= 100 OR 1=1 #

→ এতে মিলবে 121

true শব্দই ফলা

ফালা

string মানে
একটা string

SQL injection prevent করা জন্য

bind parameter use করা

→ framework এ use
করি।

→ Scenario থেকে SQL inject করা
prevent করা xm এ থাকবে না

→ Session Hijacking মানে হলো
এই method.

ଏହି class (ଆଉ (web security) ଏକଟି ଶ୍ରେଣୀ ଅଟେ) ^{**}

class note from tonu:

BooTforce attack ?

↳ prevention ଏବଂ way କି ?
(2FA / multifactor)

SQL injection

↳ framework
↳ bind parameter

→ ଦିଅ comment

in url: login.php

Site: .in/.bd/.pk

Filetype: pdf

Wayback machine

Done

↳ advance
search key
word

ଅନ୍ୟତମ keyword
advance searching
↳ କାହିଁକି ତାହା
done ବୋଲି,
open source
Intelligent,

→ SQL injection କରାଯାଇ କାହା

→ bind parameter କରାଯାଇ କାହା

session hijacking

HTML:

html List:

* Ordered list:

`<ol type="a">`
↓
a जिस Serial
है a, b, c

`<ol>`

`<li> P </li>`

`<li> Q </li>`

`<li> R </li>`

`</ol>`

1. P

2. Q

3. R

*

* Unordered list:

`<ul type="square">`
↓
□ □ □ □

`<ul>`

`<li> W </li>`

`<li> D </li>`

`<li> M </li>`

`</ul>`

• W

• D

• M

* Description list:

`<dl>`

`<dt> Front </dt>`

`<dd> The port # </dd>`

`<dt> Back </dt>`

`<dd> The server </dd>`

`</dl>`

Front

The port #

Back

The server

Nested:

 Fruits

 A

 B

 veg

• Fruits

• Apple

• B

• veg

* HTML Form:

First Name :

input type = text

Last Name :

Password :

input type = "password"

Gender :

☐ Male ☐ Female ☐ Others

Radio button

Religion :

☐ Islam ☐ Hindu ☐ Others

checkbox

Dept. :

dropdown menu

Comment

text area

Reset

Submit

submit, Reset button

<input type="button">

<input type="checkbox">

<input type="color">

<input type="email">

<input type="password">

<input type="radio">

<input type="submit">

<input type="text">

ex:

<body>

<h2> Registration form </h2>

<form>

Fullname: <input type="text">

or

<label> Fullname : </label> <input type="text">

or

<label for="fullname"> Full Name : </label>

<input type="text" name="fullname" id="fullname">

</form>

Fullname:

<input type="text"

placeholder="Enter name">

Submit button <input type="submit">

number input:

<input type="number" max="100" min="0">

↳ 0 to 100 পর্যন্ত input

দিয়ে দেবে।

*. Select:

<select>

<option value= - >

<option value= - >

</select>

*. Text Area:

<textarea cols="30" row="10">

</Hextarea>

*. Image tag:

<img src

= "bd. jpg">

↳ same folder নামক স্থানে
নাম রাখতে হবে।

`<img src = " " width = "400" height = "200" >` ^{auto}

↳ size control

* object tag: (आ) कान file view करण (आ)

`<object src = "hello.txt" > </object>`

↳ type बूट करण मितो मितो

`<embed src = "m.mp4" type = video/mp4 >`

* Anchor/link:

`<a href = "https://google.com" > Visit Google </a>`

visit google show करण एकर एकर link
click करण given link एनिम शरण।

` Open in
New Tab ` ननून tab ए open शरण।

↳ click करण

↳ making an Image Clickable:

`
<img src = "logo.png" alt = company
logo >
`

* HTML block element : Takes full width.

Starts from new line.

`<div> </div>`

`<h1> </h1>`

`<p> </p>`

`<ol>, <ul>, <li>`

ex: `<div> hello world </div>`

`<p> apple </p>`

hello world

apple

এককম full block নিম্ন নিম্ন।
~~`<div> hello world </div>`~~

* HTML Inline element : Takes only

required width. Does not start new line.

`<span> </span>`

`<a> </a>`

`<b> </b>`

`
Hello world `

`<a> Hello world </a>`

* HTML block element: Takes full width.

Starts from new line.

`<div> </div>`

`<h1> </h1>`

`<p> </p>`

`<ol>, <ul>, <li>`

ex: `<div> hello world </div>`

`<p> apple </p>`

hello world

apple

প্রকৃত full block নিম্ন নিচে।
~~`<div> hello world </div>`~~

* HTML Inline element: Takes only

required width. Does not start new line.

`<span> </span>`

`<a> </a>`

`<b> </b>`

`
Hello world `

`<a> Hello world </a>`

`<span>green </span>`

↳ Hello world green

↳ ଏହା ଦେଖା ଗୁଡିକ (ସବୁକିଛି)
କାମକା କାମ କରୁଥିବ
ନିଶ୍ଚୟ ।



Block element vs Inline Element

- block
- Takes full width
- New line starts
- Ex: `div`

- inline
- Takes needed width
- Same line stays
- Ex: `span`

* html version:

The current version of HTML is the 'HTML Living Standard' which is continuously updated by the web WHATWG.

- Modern html uses descriptive tags like `<header>`, `<footer>`, `<article>`, and `<nav>`
- We can embed video and audio directly without needing

third party plugings like Flash

→ Simplified Doctype: `<!DOCTYPE html>`

* HTML basic structure:

`<!DOCTYPE html>` → browser knows what type of documents to expect.

`<HTML>`

`<HEAD>`

`<title>`

`</title>`

`</HEAD>`

→ title type info.

`<BODY>`

CONTENT

→ web page ki
ki prdakti
hota hai

`</BODY>`

`</HTML>`

* html body tag:

The `<body>` tag defines the documents body. The `<body>` element contains all the contents of an HTML document, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.

Note-1: `<body>` elements must be below the `<head>` elements

Note-2: There can only be one `<body>` element in an HTML documents.

`<head>`

`</head>`

`<body>`

`<h1> hello </h1>`

`<p> This is web </p>`

`</body>`

`style = "color: red;"`

→ body is not red color

background: black;

↓
background black

* header tag:- Used for website header section

Usually a company logo, the main website title, may be a search bar, and very often the navigation menu.

```

<body>
  <header>
    <div>Help Operation</div>
  </header>
  <nav>
    <ul>
      <li><a href="#">Home</a></li>
      <li><a href="#">About</a></li>
      <li><a href="#">Contact</a></li>
    </ul>
  </nav>
</body>

```

navigation menu

* <div> creates a box for styling, while <header> creates a box with meaning.

head vs header:

| | |
|--|-------------------------------------|
| Hidden, inside <html> metadata, <css>, <title> | Visible, inside <body> (Logo, menu) |
|--|-------------------------------------|

div vs header:

| div | header |
|---------------------------------|--------------------|
| → No meaning | → has meaning |
| → Used for layout & CSS styling | → Used for content |
| → Google ignores it | → Google uses it |
| → A Generic box | → A container |

* div tag: Container

```

Used to divide sections
<body>
  <div style="border: 1px solid black; padding: 5px;">
    <div>
      <div>
        <div>
          <div>
            </div>
          </div>
        </div>
      </div>
    </div>
  </body>

```

User -> Layout design, then Apply CSS.

head vs header.

hidden, inside <html>
metadata, (css, title)

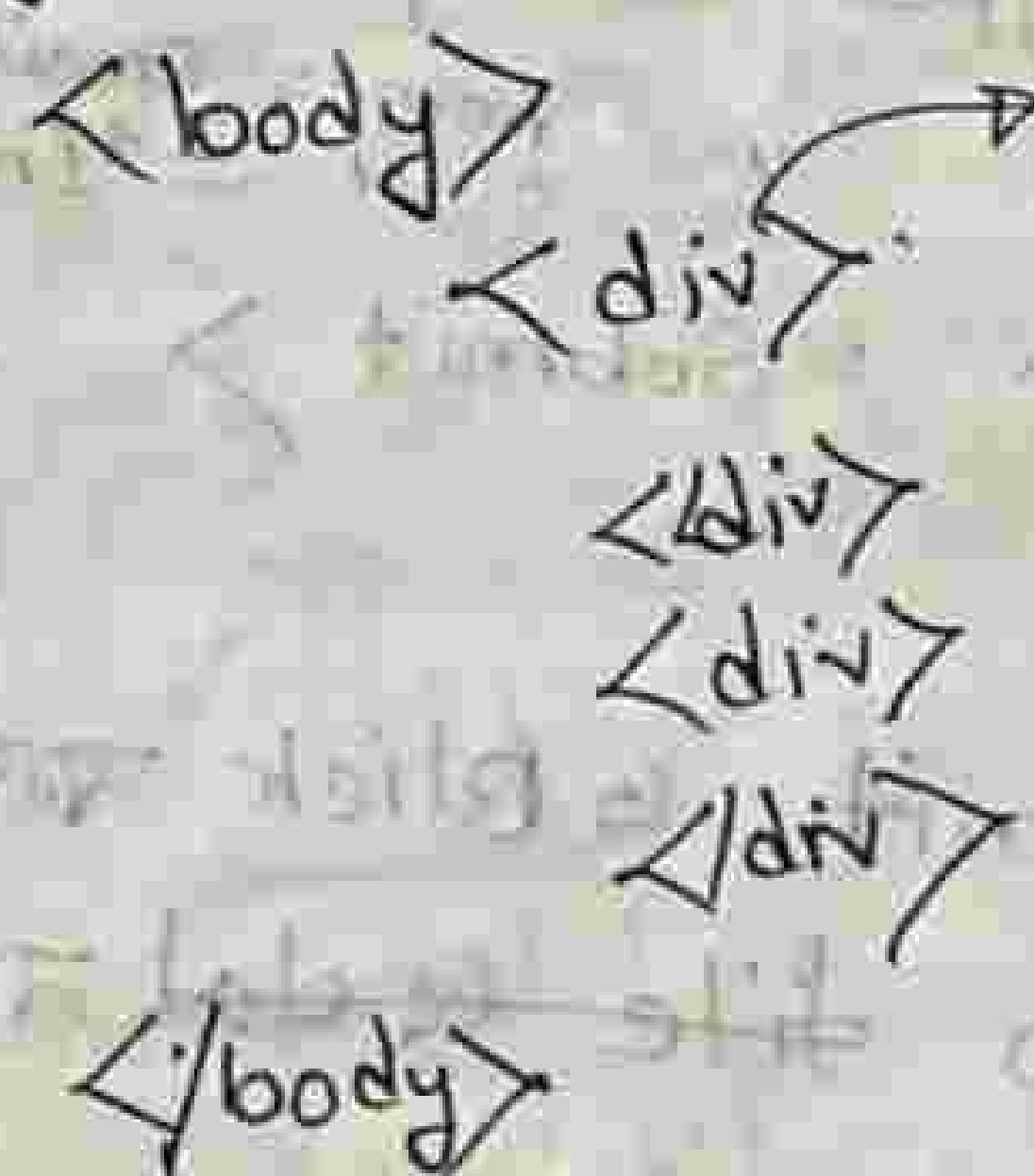
visible, inside <body>, page heading
(Logo, menu, heading)

div vs header:

| div | header |
|---------------------------------|--|
| → No meaning | → has meaning |
| → Used for layout & CSS styling | → Used for grouping the website logo, title, & navigation menu |
| → Google ignores it | → Google reads it |
| → A Generic box | → A labeled box. |

* div tag: Container

used to divide sections



style = "border: 1px solid black"
↳ border color, width, style
or div &

div এর ডিভিও সব
সামান্য চালাবে

Uses -

→ Layout design, Group elements,
Apply CSS.

*. Form and input Get (post method) :

<h1> Form </h1>

<form>

Name : <input type="text">

Email : <input type="text">

Submit : <input type="submit">

</form>

<form action="X.php" method="get">

Name : <input type="text" name="Name" placeholder="Name">

Email : <input type="text" name="Email" placeholder="Email">

Submit : <input type="submit" value="Submit">

</form>

↳ Submit & Click कराने

x.php file को नामक file

data मा

x.php:

<h1> Form Data Action </h1>

<?php

<button type="submit" value="Submit">

* headings:

`<h1> A </h1>`

`<h2> B </h2>`

`<h6> C </h6>`

h1 = biggest heading

h6 = smallest heading

Imp't Qstn: How do i show h6 big &

h1 small?

Ans: Using inline css,

`<h6 style="font-size: 50px;"> A </h6>`

`<h1 style="font-size: 12px;"> B </h1>`

* html Favicon: to tab a image show

`<head>`

`<title> HTML Favicon </title>`

`</head>`

`<body>`

`<h1> A BC </h1>`

`</body>`

`<link rel="shortcut icon"`

`href="x.png" type="image"`

↳ image location

↳ add a .png image + HTML

Favicon (फविकन)

html = structure, CSS = beauty / design

CSS (Cascading Style sheet)

* Basic Syntax:

selector → property name

color : green;

property value

p/h

याहू
design करण
आम निम

p {

color : green;

font-size = 20px;

}

ex:

html file:

<body>

<p> A </p>

<h3> B </h3>

<p> c </p>

</body>

CSS file:

h3 {

color : red;

}

h3 को color (मथार
red

p {

background-color : yellow;

font-size : 25px;

}

background
yellow थार
both p को
करे वर
size 25px
हो।

class:

<body>

<h2> A </h2>

<h3> B </h3>

<h4> C </h4>

<h2 class = 'ab' > A </h2>

<h3 class = "ab" > B </h3>

<h4 class = "ab" > C </h4>

কিভাবে কাজ করে,

~~সবচেয়ে~~ so

</body>

<style>

.ab {

color : red;

</style>

→ class declare =

• class name

আনেকগুলো element same
চরিত্রের জন্যে class use করি,
এখানে h2, h3 & h4 এর color
সব red হবে।

আনেকগুলো element এ same style apply
করলে class

* id:

→ more priority than class

<body>

<h2 id="xy">A </h2>

<h2 id="xy">B </h2>

<h2>C </h2>

</body>

style:

<style>

h2 {

color: red;

#xy {

color: green;

background-color: yellow;

ଉତ୍ତର,

id = xy ଏବଂ ଏହା ଚାହୁଁଥିବା
green ଓ background yellow
ହେବ। କିନ୍ତୁ ଚାହୁଁଥିବା red
ନାହିଁ।

<h2 id="xy" class="ab">T </h2>

କେବଳ particular
single element ଓ
style apply ହେବ।
id ବ୍ୟବହାର କରାଯାଏ

| Feature | class | id |
|----------|-------------------|-----------------------|
| Symbol | . | # |
| Used for | Multiple elements | single unique element |
| Reusable | Yes | No |
| Syntax | . box | # header |
| priority | lower | higher |

<div>

.ab {

border: 2px solid black;

ଏହାଦ୍ୱାରା
ବର୍ଗାକାର border
ଲାଗେବ।

* inline CSS:

html file ko

<body>

<h1 style="color: red;">Title 1</h1>

<p style="width: 250px; color: orange;">

ABCDEF </p>

<h1 style="color: blue;">Title 2</h1>

<p> ABCDEFG </p>

</body>

add karke dikhiye
-> jab (M) <h1> ko style
add karke dikhiye inline CSS.

output:

Title 1 -> red

ABCDEF -> 250px + orange (M)

Title 2 -> blue (M)

ABCDEFG < -> black (by default)

x. internal CSS: same html file में `<style>`

tag के साथ CSS apply करना है,

index.html:

`<style>`

`h1 { color: blue; }`

`.+ { color: red; }`

`# info { background: orange; }`

`</style>`

`<body>`

`<h1> Title 1 </h1>`

`<p class = "+" > ABCDE </p>`

~~`<h1>`~~ `<h1 class = "+" > Title 2 </h1>`

`<p> ABC </p>`

`<h1> Title 3 </h1>`

`<p id = "info" > ABCDEFG </p>`

`</body>`

output:

Title 1 → blue

ABCDE → .+ class ko property me red

Title 2 → red (दस्तावेज)

ABC → black (by default)

Title 3 → Blue

ABCDEFG → background orange (दस्तावेज)

* External CSS: ଏହା external style CSS

file ଓ style design କରା ହେବ,

style.css:

```
h1 {color: blue;}
p {color: red;}
#intro {background: orange;}
```

index.html

previous ଏବଂ ଅନ୍ୟ
just <style> tag
ଏବଂ value ଥିବ
ଏବଂ style.css
ଏବଂ ଏହା ଏବଂ
connect କର
ଦିଅନ୍ତୁ,
<style> → tag
ବାଦ,

```
<link rel="stylesheet"
href="style.css">
```

↳ କୁଡ଼ିଫାଇଲ

Output:

same

name ସଫି
folder name
ହେବ

priority: inline css > Internal css > External css

class vs id

| class | id |
|-------------------------|---------------------------------|
| → CSS Symbol: . (dot) | → # (hashtag) |
| → Many times per page. | → Only once per page |
| → Low priority | → High priority |
| → A uniform for a group | → A fingerprint for one person. |
| → Reusable | → Unique. |

→ determines which CSS rule will apply when multiple rules target the same element.

* CSS specificity: কোন CSS কাজ

করবে সেটা নির্দিষ্ট করে দেওয়া। priority basis
(এ কাজ করে।)

যদি এমন হয় যে inline, internal, external
আলাদা আলাদা style দিতে চাচ্ছে একই line-এ
তখন inline-এর priority বেশি তাই
এটা accept করবে।

inline > internal > external → এভাবে কাজ করে।

css file
color in CSS:

style. CSS:

+

color: rgb(255, 0, 0);

→ red color customized value দিও
অন্য। A

/* AB */ → comments

color: rgba(red, green, blue, alpha);
→ value দিও

alpha = 0 to 1

background color: rgb(, ,)

height: auto

→ যতটুকু আছে নিয়মিত

min-height:

→ minimum বাকু
অন্য must

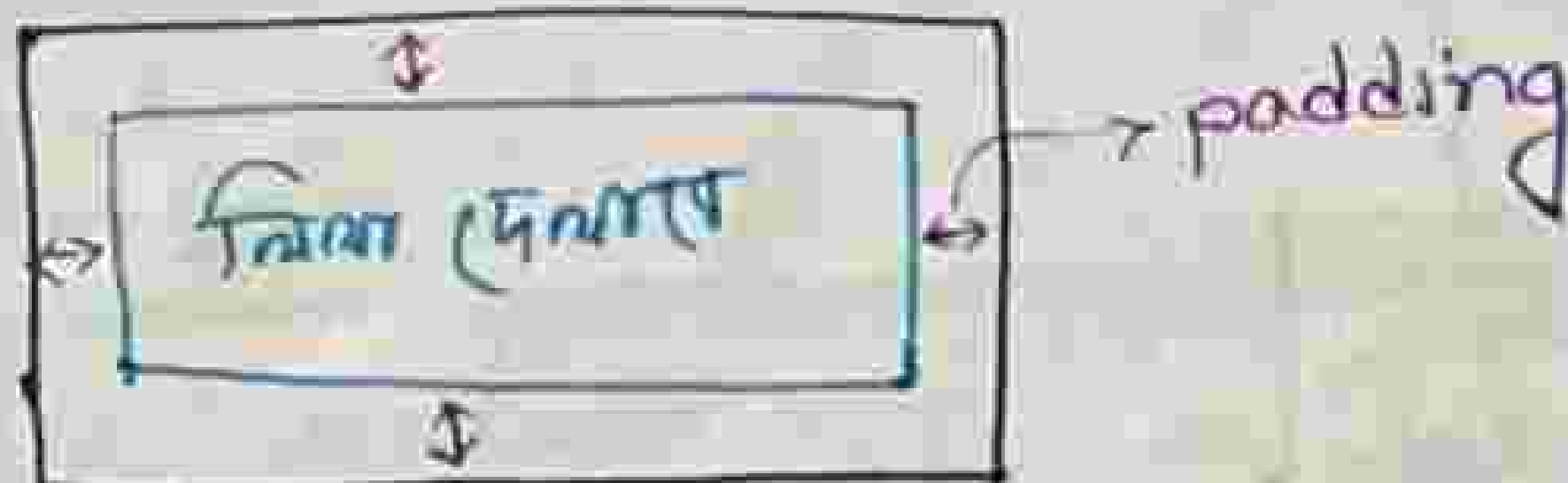
max - :

→ max যতটুকু হতে পারে

width: auto

→ যতটুকু আছে

* CSS padding: inside space



index.html
`<link rel="stylesheet" href="style.css">`
`<body>`

`<div class="box">Lorem -`

`</div>`

`</body>`

style.css

.box {

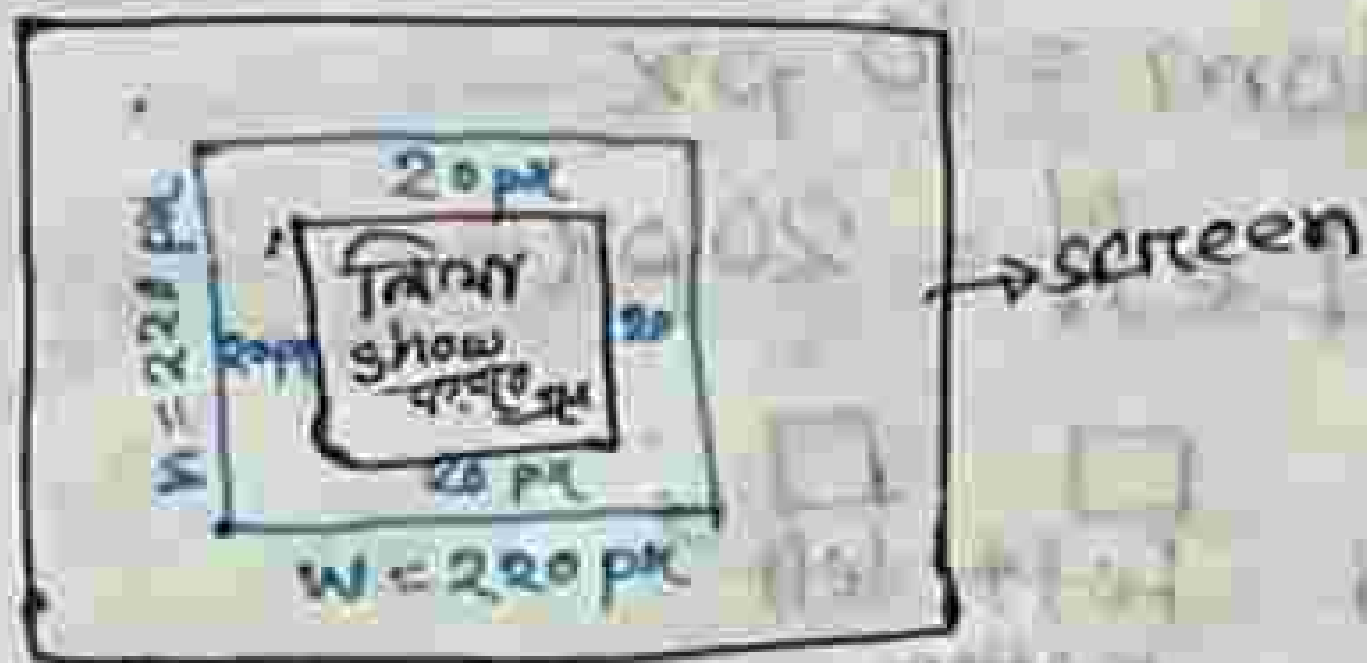
background-color: white;

height: 220px;

width: 220px;

padding: 20px

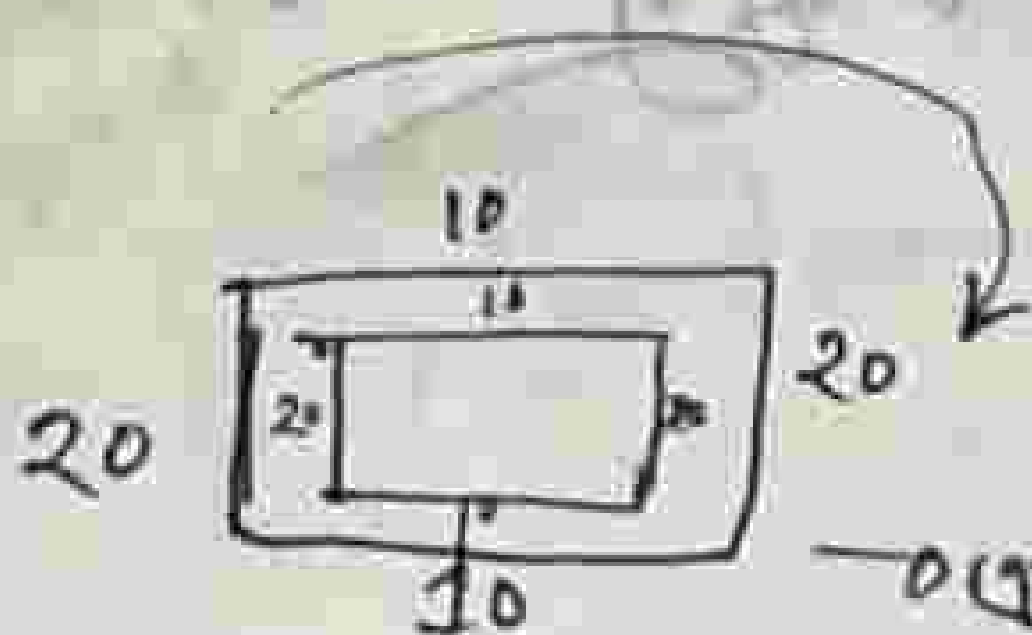
Output:



padding-left: 100px

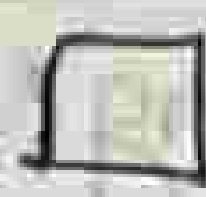
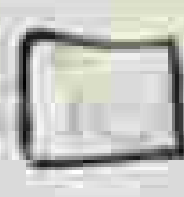
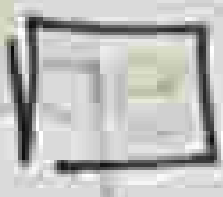
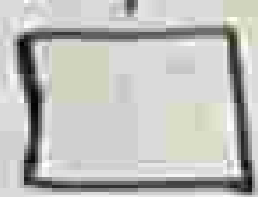
↳ ଡାହାଣ Left (ଆମେ 100 px ଆମେ ଡାହାଣ, ଡାହାଣ right, top, bottom ଓ ଡାହାଣ କରାଯିବ)

ଯଦି, padding: 10px 20px; ଦୁଇ value (ଦିଶାରେ)



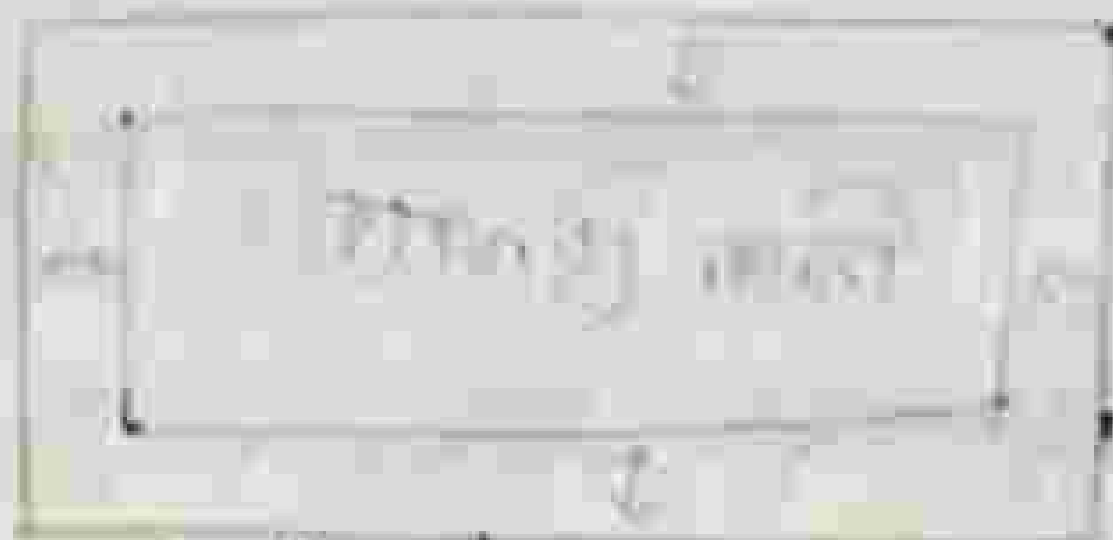
ଠାରେ (ମଧ୍ୟ) ଥାଏ [top & bottom = 10px
left & right = 20px]

padding : ^{top} ^{Right} ^{bottom} ^{Left}



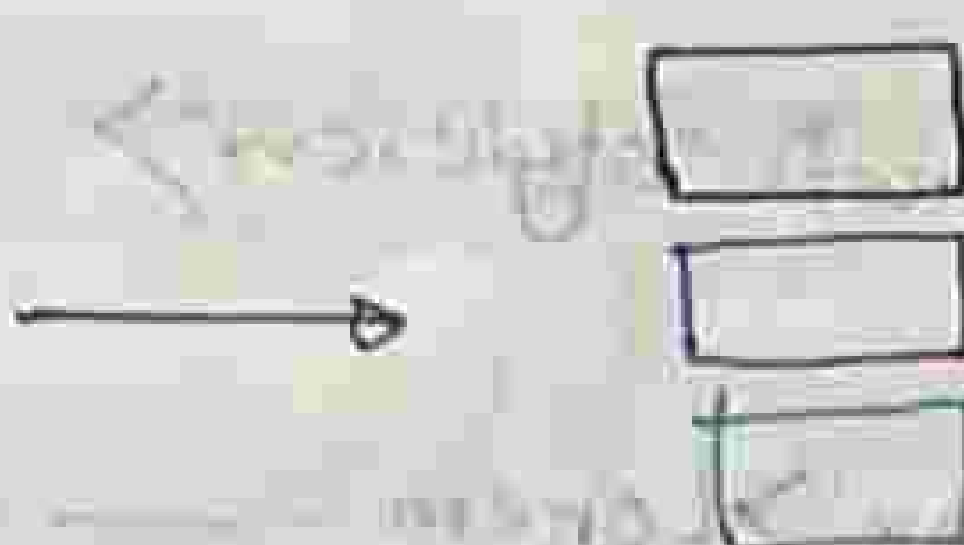
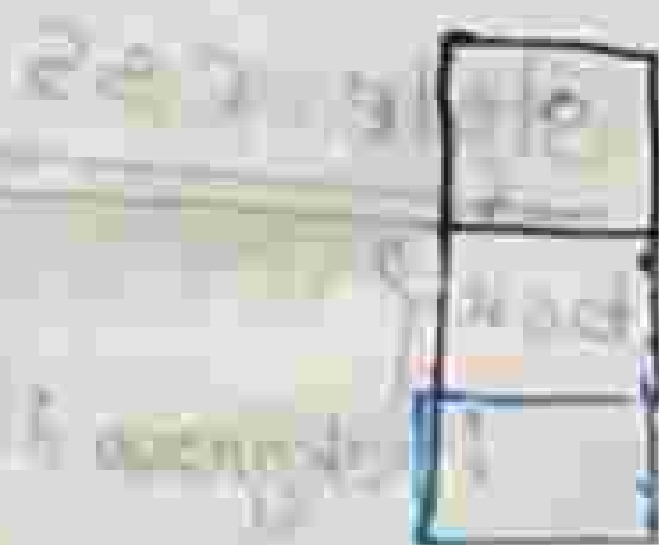
২৪টা ভিন্ন ভিন্ন value
দিতে পারবে।

padding



* CSS Margin :

Outside space



padding এক মতো করেই ঘটিবে কাঠের দিকে কাজ
করে।

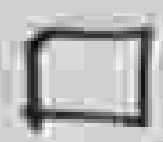
margin : 10px → সব দিকে 10 কাজ হবে

margin : 10px 20px

↓
top = bottom = 10px

Right = Left = 20px

Margin :



top

Right

left

bottom

margin-left : → শুধু left side



margin left & Right = auto ~~का~~ ~~दिए~~ ~~अब~~ ~~मध्य~~ ~~के~~ ~~centre~~
 में आयेगा।

index.html

```
<body>
<div class="b1"></div>
<div class="b2"></div>
<div class="b3"></div>
```

style.css

```
.b1 {
height: 120px;
width: 120px;
background-color: black;
margin: 20px;
}

.b2 {
height: 120px;
width: 120px;
background-color: black;
margin: 20px;
}

.b3 {
height: 120px;
width: 120px;
background-color: black;
margin: 20px;
}
```

* Border Radius:

top & 10px & blue border border-top: 10px solid blue;
 right & red " - right: 10px " red
 bottom & orange " - bottom: 10px " orange
 left & green " - Left: 10px " green;

~~border~~ border-radius: 50px



एक कर्वा (radius)
 है।

border-radius: 50% → (circle)

* Padding vs margin

| | |
|--|---|
| → Feature: Padding | → Feature: margin |
| → Inside the border | → Outside the border |
| → Space between the content and its own border | → Space between this box and other neighboring boxes. |
| → Background color is element's background color | → No background color. |
| → Cannot be set to auto | → Can be set to auto |

* Border Radius

border-top: 1px solid black;

border-right: 1px solid black;

border-bottom: 1px solid black;

border-left: 1px solid black;

border-radius: 50%;



→ a collection of that m and w

Bootstrap with

What? → It is a front e

Why do we need Bootstrap

- Faster web de
- Easier
- Create respon

* What is sm, md

Ans:

sm = small device

md = medium devi

lg = large device

ex: <div class = "col-sm-12">

small screens

medium

md-6 sm-4 apply

<div class = "col-sm-6 col-md-4">

col-sm 4 :
on small serc

This mea

→ a collection of pre-made CSS styles that makes website look good and work on all devices without writing much CSS.

Bootstrap

what? → It is a front end framework.

Why do we need Bootstrap?

- Faster web development
- Easier
- Create responsive designs easily

* What is sm, md in Bootstrap?

Ans:

sm = small device → > 576 px

md = medium device → > 768 px

lg = large device → > 992 px

ex: `<div class = "col-sm-6 col-md-4">`

small screens → it will take 6 columns

Medium

→ " " " 4 "

md-6 & sm-4 apply concept:

~~<div class = "col-sm-6 col-md-4">~~

col-sm 4 :

on small screens,

$$12 \div 4 = 3$$

This means 3 items can fit in one row.

bootstrap files is hosted on a server somewhere else we just link to them

col-md-6, on Medium screens:

$$12 \div 6 = 2$$

This means 2 items can fit in one row.

* CDN = Content Delivery Network :

Advantage

- very fast setup
- No need to download files
- Faster loading because of server cache
- Beginner-friendly
- Works anywhere with internet connection.

Disadvantage

- Does not work without internet
- If CDN server is down, Bootstrap may not load.
- Difficult to customize

* Locally import : means downloading the bootstrap file into your own project.

ex: `<link rel="stylesheet" href="css/bootstrap.min.css">`

| Adv | Disadv |
|--|---|
| <ul style="list-style-type: none"> → Works offline → Full control over files → Easy to customize → No dependency on ^{CDN} | <ul style="list-style-type: none"> → Setup process is longer → Need to manage files manually → Project size becomes larger |

* Globally import: ^{→ CDN} means importing Bootstrap of glob. once and using it ~~thruout~~ throughout the whole project.

| Adv | Disadv |
|---|---|
| <ul style="list-style-type: none"> → No need to import repeatedly → Cleaner code → Easier management for large project | <ul style="list-style-type: none"> → Unused CSS may load → Performance may decrease slightly → CSS conflicts can occur |

* How columns divided in Bootstrap:

Bootstrap grid system contains a total of:

12 columns

ex-

```
<div class="row">
```

```
<div class="col-6"> A </div>
```

```
<div class=""> B </div>
```

```
</div>
```

↳ divided into two parts

col-12 → Full row

col-6 → Half

col-4 → One third

col-3 → one fourth



↳ 12 grid

→ 12 columns

* php ত একটি table এর গঠন (structure)

Students

| id | Name |
|----|-------|
| 1 | Ehsan |
| 2 | Rahim |

Results

| st_id | cgpa |
|-------|------|
| 1 | 3.80 |
| 2 | 3.50 |

Select students.name, Results.cgpa From Students
INNER JOIN Results ON
students.id = Results.st_id;

<?php

Connection

\$sql =

\$Res = mysqli_query(

while(\$row = mysqli_fetch_assoc(\$Res))

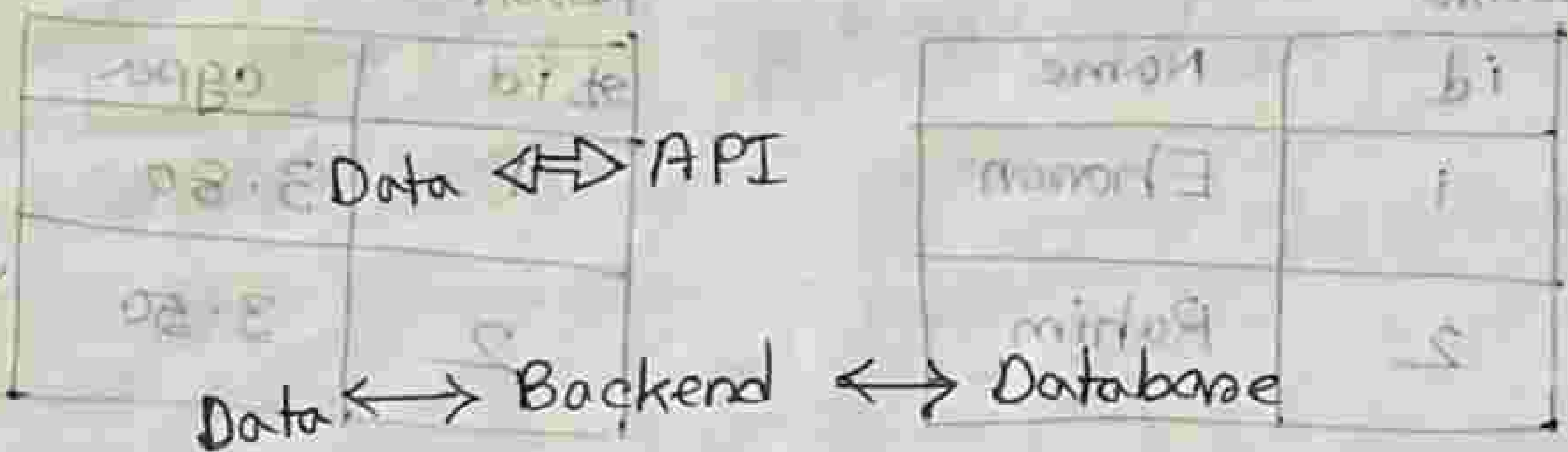
echo \$row['name'] . " - " . \$row['cgpa'] . "
";

{

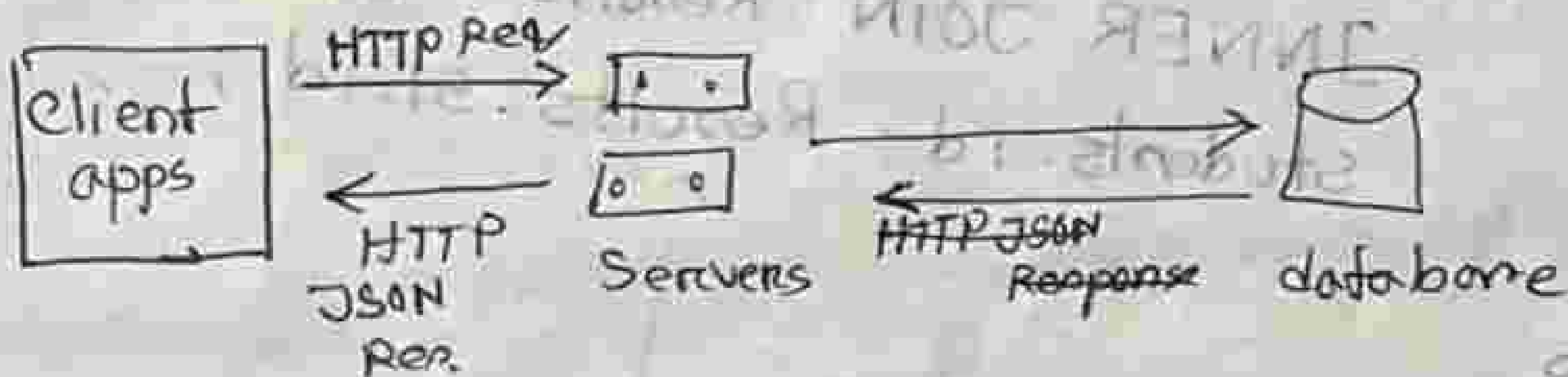
?>

;

* API: Application Programming Interface



How data works?



How to use:

Rest API

<? php

`$base_url = "https://codeforces.com/api/user-rating?";`

`$handle = "SaraJattHossain";`

`$url = $base_url . $handle;`

`$response = file_get_contents($url);`

`$data = json_decode($response, true);`

`echo $data["status"];`

API php sends
Req and
receive
data

converts json
into
php array


```

foreach($data["result"] as $result) {
    echo "<h1>". $result["contentId"]. "</h1>";
    " " . " " ["contestName"] . " " ;
    " " . " " ["rank"] . " " ;
    echo "<br>";
}

```

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Steps to use Api:

1. GET API URL
2. Send Request to the API
3. Receive Response data
4. Convert JSON data
5. Display the data

API is a system that allows communication between applications

Mostly used API format = JSON

* JSON = JavaScript Object Notation
[Used to exchange data between server and client]

* PHP → JSON (JSON encode):

function = json_encode()

ex:

```
<?php
```

```
$data = array(
```

```
    "name" => "Eharon",
```

```
    "dept" => "CSE";
```

```
    "age" => 22
```

```
);
```

```
$jsonData = json_encode($data);
```

```
echo $jsonData;
```

```
?>
```

Output:

```
{ "name": "Eharon", "dept": "CSE", "age": 22 }
```

It converts php array/object into JSON format.

* JSON Decode: It converts JSON data into php array/object.

ex:

```
<?php
$json = '{"name": "Ehanan", "dept": "CSE",
        "age": 22}';
$data = json_decode($json, true);
echo $data["name"];
```

?>
output: Ehanan

Setter, Getter, Constructor in Java

User.java:

```
public class User {
```

```
    private String name;
```

```
    private int age;
```

```
    public User() {
```

```
    }
```

```
    public User(String name, int age) {
```

```
        this.name = name;
```

```
        this.age = age;
```

```
    }
```

```
    public void setName(String name) {
```

```
        this.name = name;
```

```
    }
```

```
    public void setAge(int age) {
```

```
        this.age = age;
```

```
    public String getName() {
```

```
        return name;
```

```
    public int getAge() {
```

```
        return age;
```

Main Java:

```
public class Main {  
    public static void main (String [] args) {  
        User user = new User();  
        User user = new User ("Eha", 22);  
        user.setName ("E");  
        user.setAge (22);  
        System.out.println (user.getName());  
        System.out.println (user.getAge());  
    }  
}
```

Example
value type
parameter
constructor
method

Constructor vs Method:

Same name as class

No return type

Runs

Automatically

Any name

Has Return type

Called manually

Hacking:

SQL injection: is a cyber-attack where a hacker tricks a website into running malicious database commands

* Brute Force attack: is a hacking method that uses guesswork to break into an account.

* How SQL injection work?

When user input is directly concatenated into SQL queries without validation. Attackers can inject SQL commands that alter query logic.

* Effect of SQL injection:

- login bypass
- Data theft
- Data modification
- Database Deletion
- Unauthorized access

injects malicious SQL code into an input field to manipulate a database

* Normal query vs injected Query:

Select * from users
where username = 'admin'
OR AND password = '1234';

Select * from users
where username = 'admin'
OR 1=1 --
AND password = '';

* OR 1=1 this condition that always evaluates true. → comments

* How can SQL injection be prevented?

- Prepared statements (bind parameter)
- Parameterized queries
- Input validation
- ORM frameworks
- Least privilege access
- blind parameter

* Prepared statement: is a secure SQL execution method where user input is separated from SQL commands.

* parameterized queries:

input is treated as data only, not

executable SQL code.

* Write a secure php login query
using prepared statements:

<?php

```
$stmt = $conn->prepare("select * From
```

```
users Where username=? AND
```

```
password=?");
```

```
$stmt->bind_param("ss", $username,  
($password));
```

```
$stmt->execute();
```

(s) → string, string → data type

* Session Hijacking: is an attack where an attacker steals a user's session id and uses it to gain unauthorized access to the system.

How work!

- User logs into a website
- Server gives a session id
- Attacker steals that session id
(via-cookies, xss, network sniffing)
- Attacker uses it to impersonate user.

Prevent: Use HTTPS, Regenerate session id,
Set Secure Cookies, Short Session
expiry